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SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 1

Aim: Getting Started with Tableau and Building Basic Visualizations

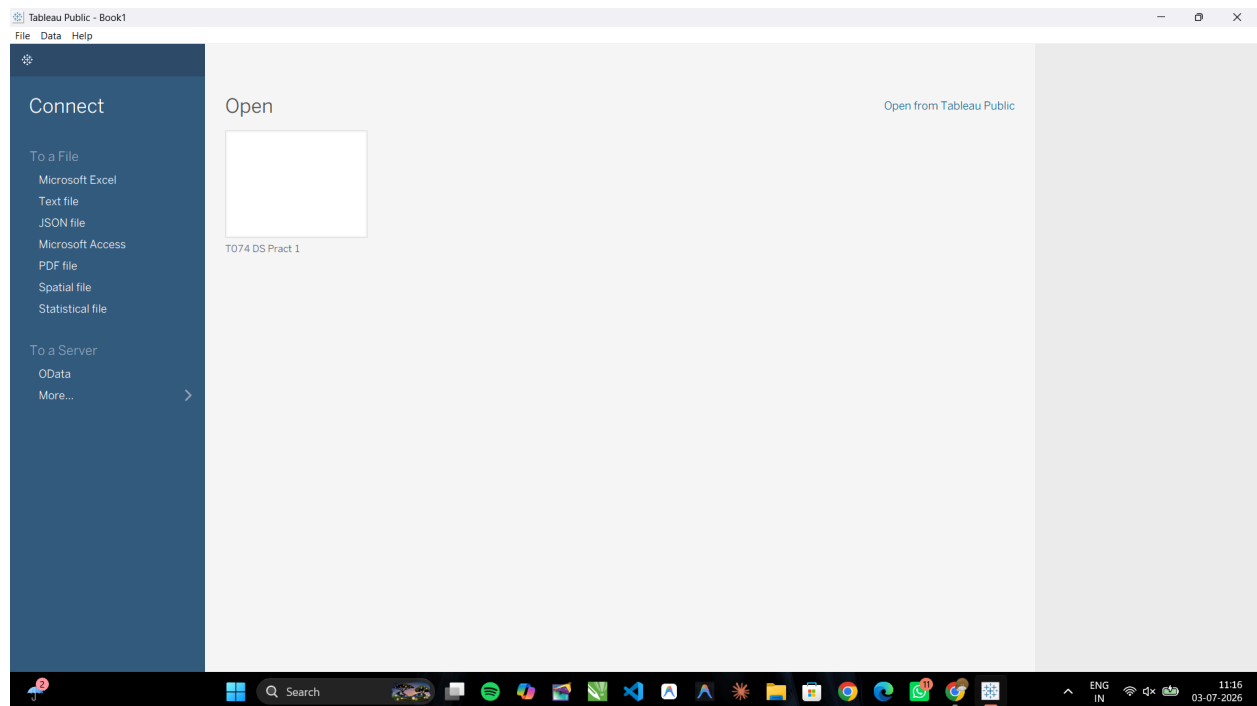
- a) Connect to a CSV dataset and explore the Tableau interface
- b) Identify and differentiate Measures and Dimensions
- c) Convert fields between discrete and continuous and observe changes
- d) Create and customize a basic bar chart
- e) Build a line chart for time-series analysis
- f) Create a geographic map visualization
- g) Combine multiple charts into a simple interactive dashboard

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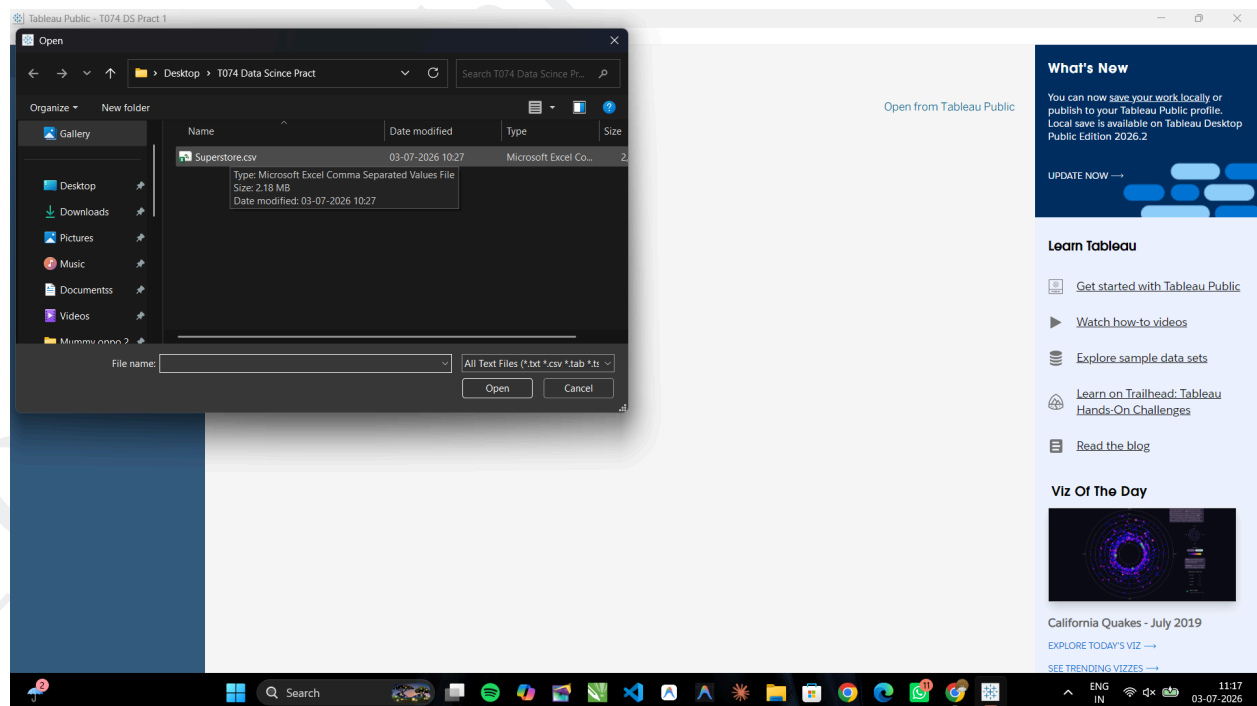
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Part A: Connect to Dataset & Explore the Interface

1. Launch Tableau Desktop on your machine



2. On the left-hand blue navigation pane under Connect, click on Text File (Tableau treats CSVs as text files)



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3. Select your CSV file from your Windows Finder and click Open

The screenshot shows the Tableau Desktop interface with the 'Superstore.csv' file loaded as a data source. The sidebar on the left shows the 'Connections' section with 'Superstore' selected, and the 'Files' section with 'Superstore.csv' listed. The main view displays the data source details for 'Superstore.csv', including a description and a list of fields. The data preview table at the bottom shows the first six rows of the data.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman

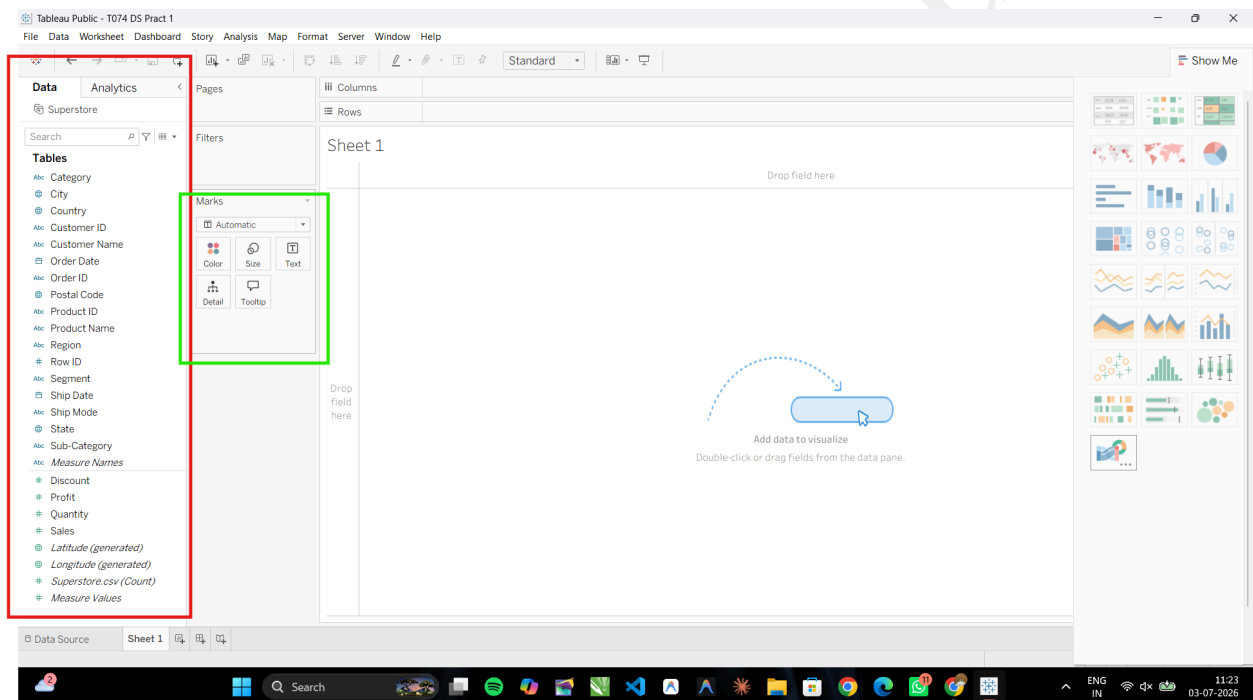
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4. You will be taken to the Data Source page. Here, preview your data rows at the bottom to ensure they look correct, then click Sheet 1 at the bottom left to enter your workspace. Tableau will bypass its faulty drivers, create a native workspace called Clipboard, and instantly open up Sheet

Navigating the Interface

- The **Data Pane** (Far Left): This holds all your variables from the dataset.
- Shelves (Top/Center): The Columns and Rows shelves act as your \$X\$ and \$Y\$ coordinates.
- The **Marks Card** (Left of center): This controls your visual design properties like color, text labels, and sizes



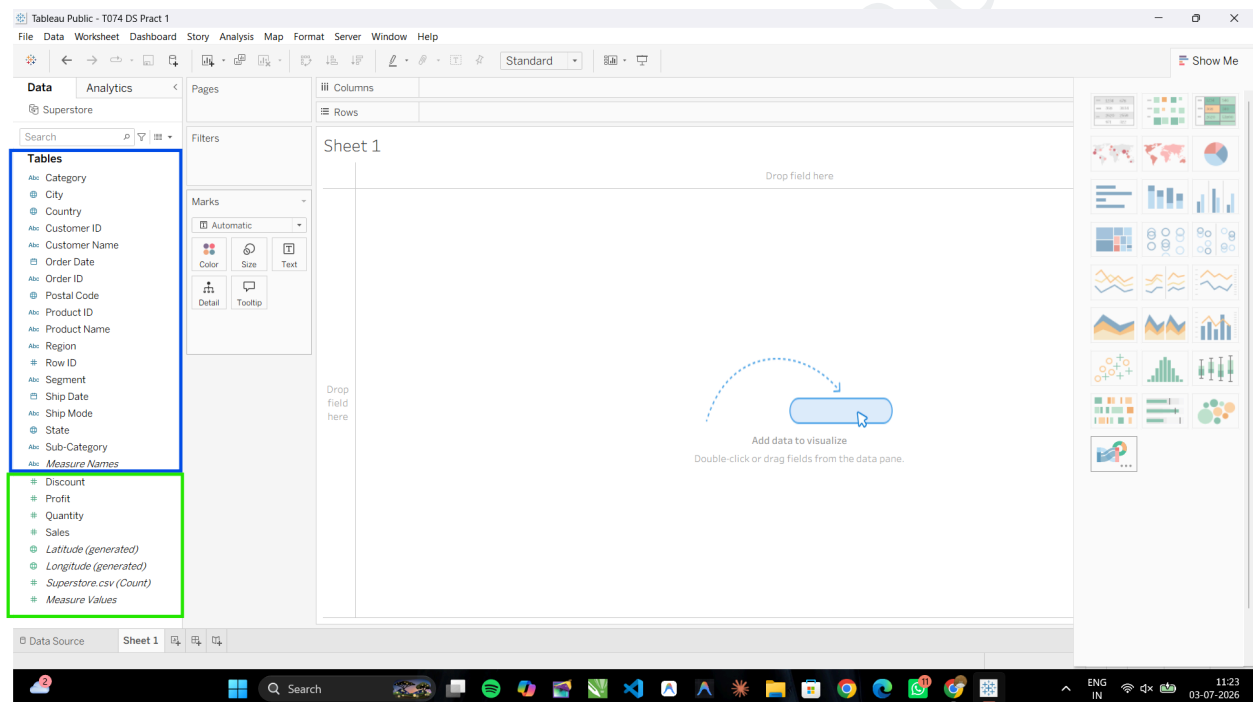
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Part B: Identify and Differentiate Measures and Dimensions

Look closely at your Data Pane on the left. Tableau reads the metadata of the Superstore CSV and splits your items automatically:

- **Dimensions** (Categorical / Blue Icons): Fields containing text, dates, or geography. In Superstore, look for fields like Category, Segment, State, or Order Date. They slice your data into distinct groups
- **Measures** (Numerical / Green Icons): Fields containing raw quantitative values. Look for fields like Sales, Profit, or Quantity. Tableau will automatically summarize or average these values whenever you place them on a chart



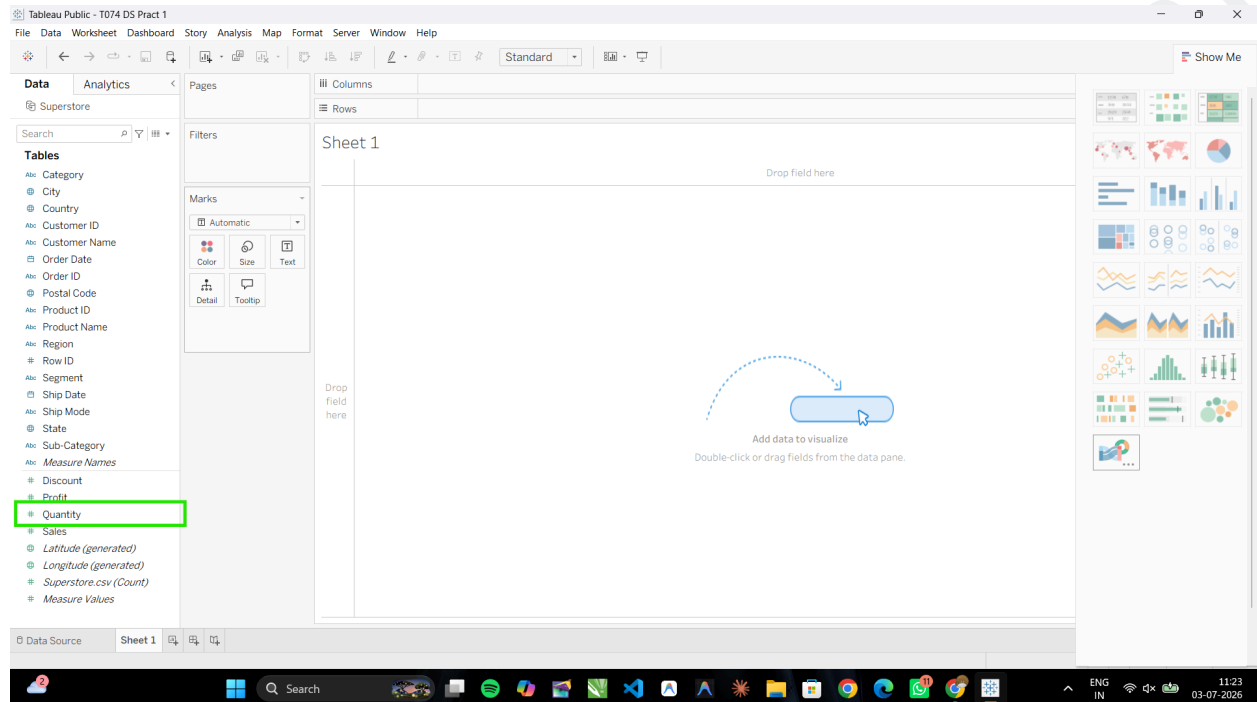
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Part C: Convert Fields Between Discrete and Continuous

Tableau colors its data pills to represent how it visualizes information. Blue means Discrete (separate header blocks) and Green means Continuous (an infinite unbroken axis)

1. Find the Quantity field in your green measures list.

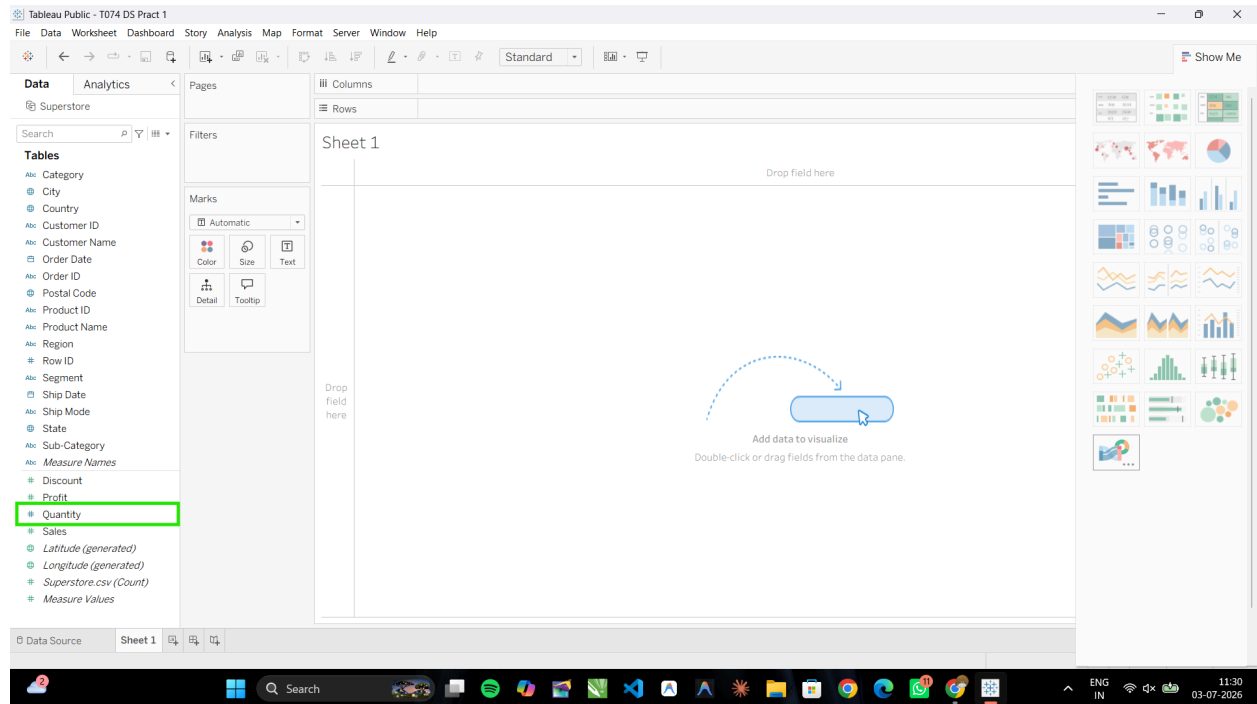


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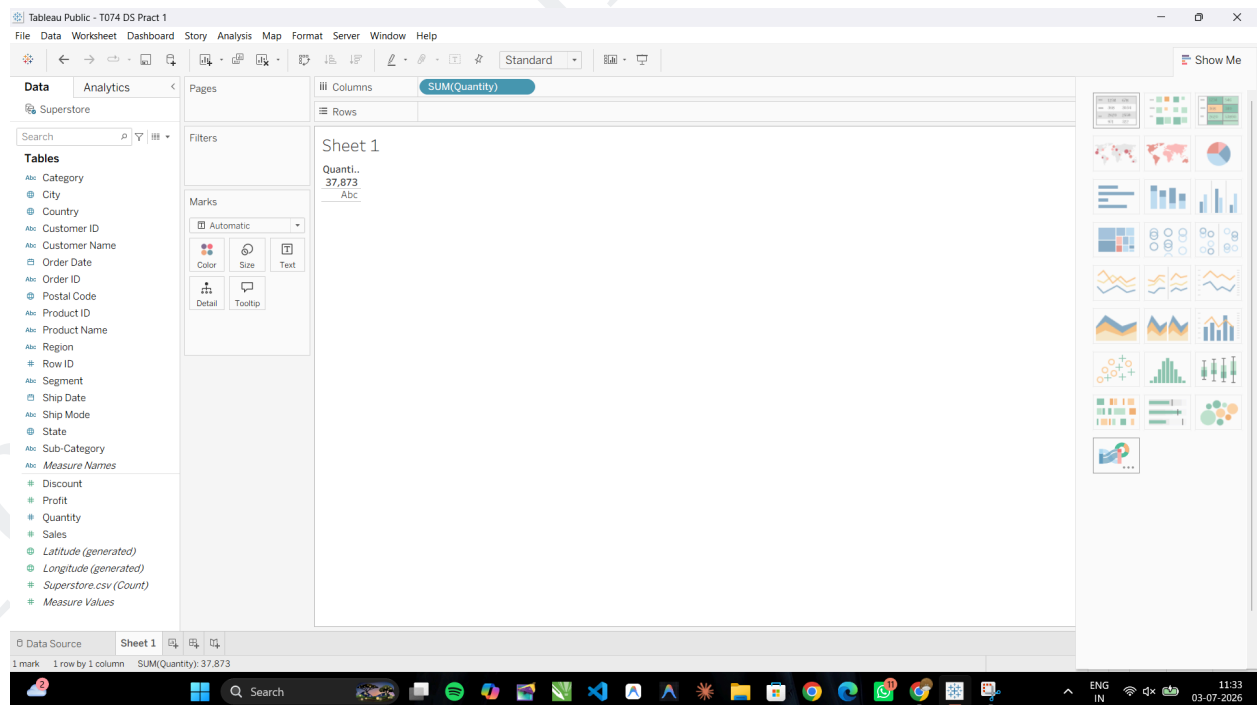
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2. Right-click (or hold Ctrl and click) on Quantity and choose Convert to Discrete.

Notice its icon changes from a green # to a blue #



3. Drag this new blue Quantity pill onto your Columns shelf. It will draw isolated individual header columns for each exact number (1, 2, 3, etc.)



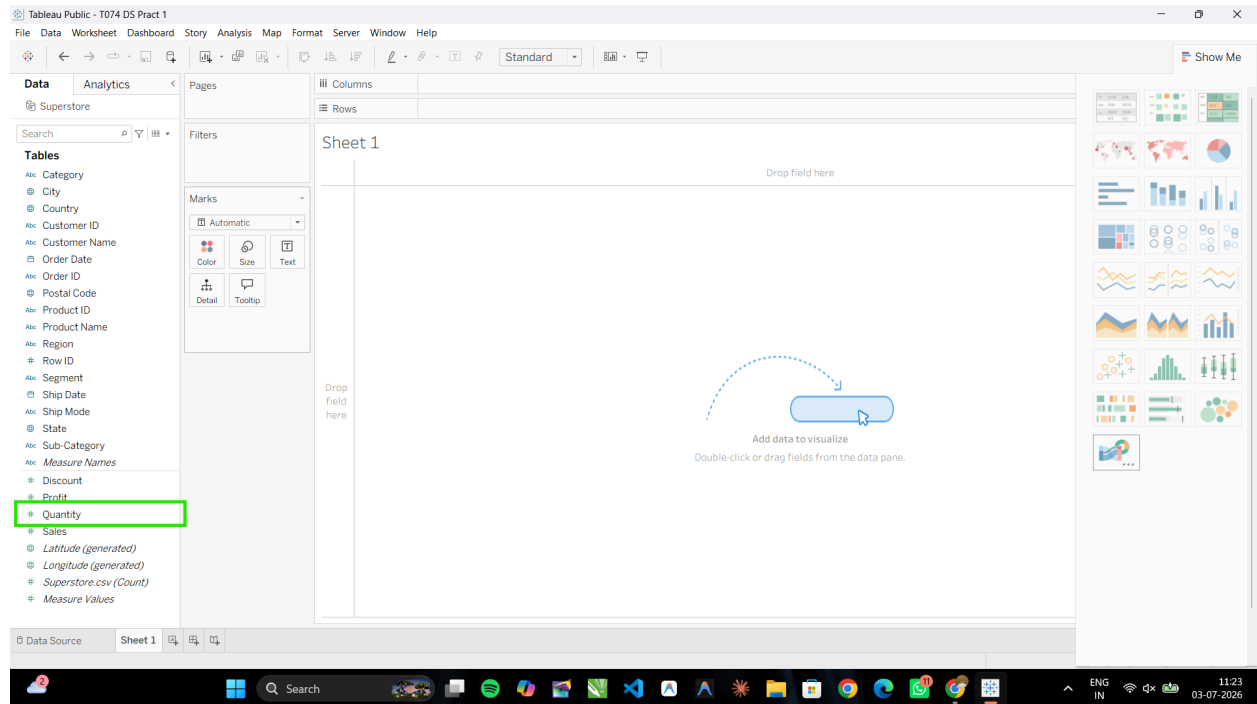
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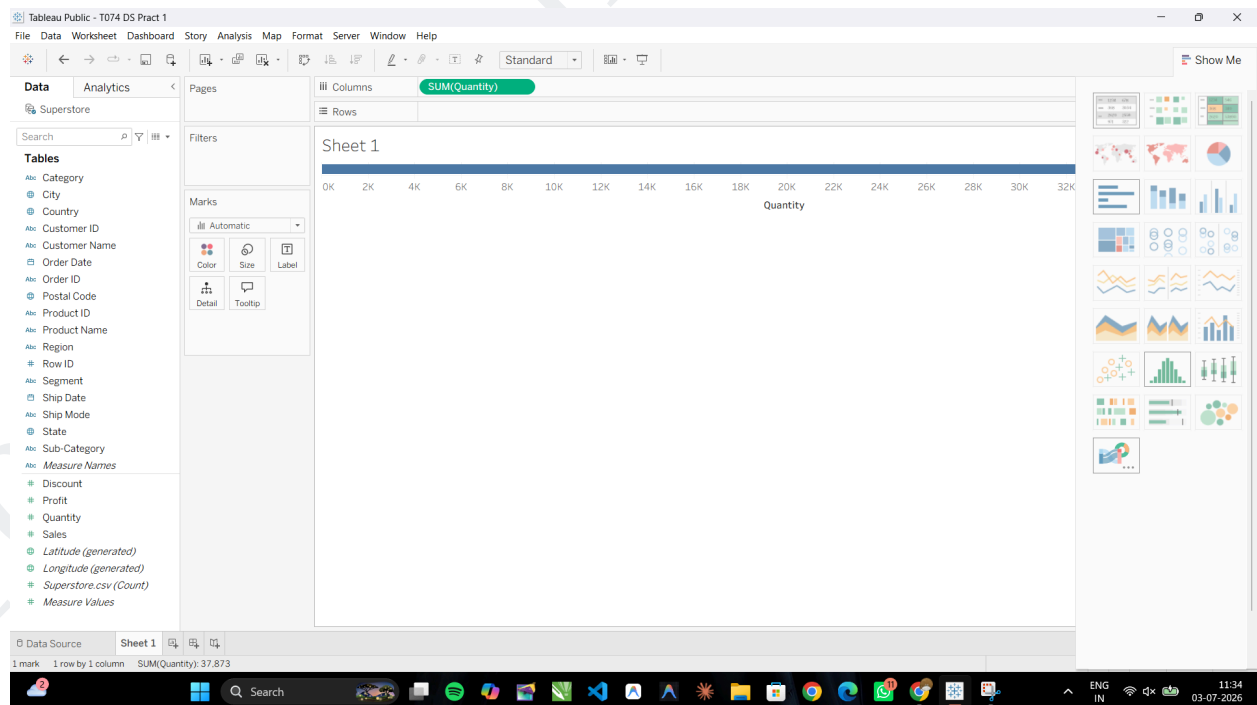
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4. Drag it back off the shelf. Right-click Quantity again and select Convert to Continuous.



5. Drag it back onto Columns. Instead of individual column blocks, Tableau now draws a singular, smooth, continuous numeric axis line.

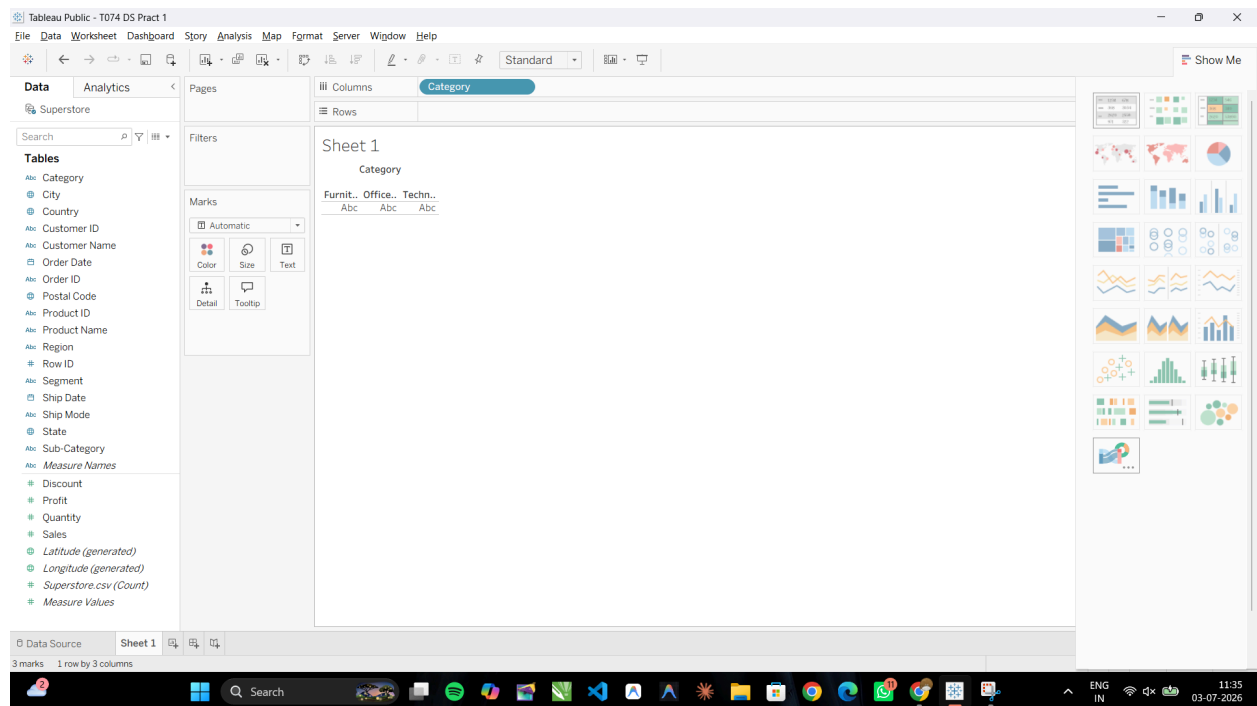


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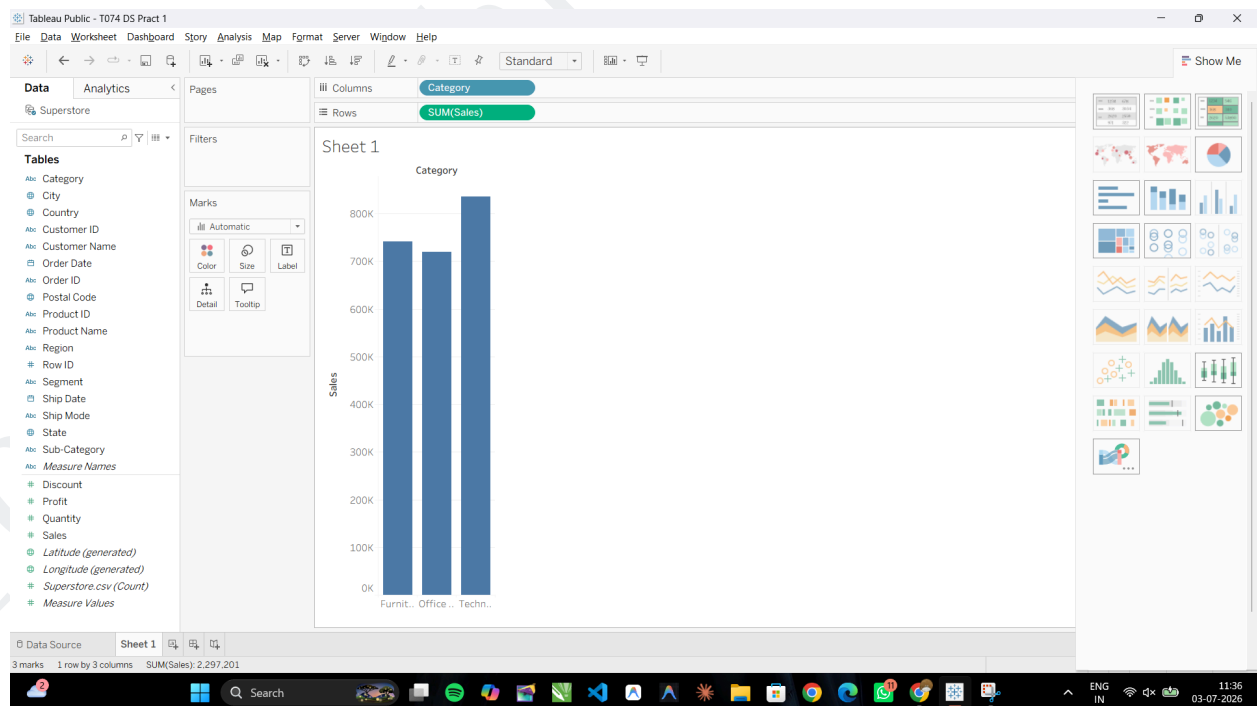
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Part D: Create and Customize a Basic Bar Chart

1. From your Data Pane, find Category and drag it up onto the Columns shelf



2. Find the Sales measure and drag it up onto the Rows shelf. You will instantly get three vertical bars representing Furniture, Office Supplies, and Technology



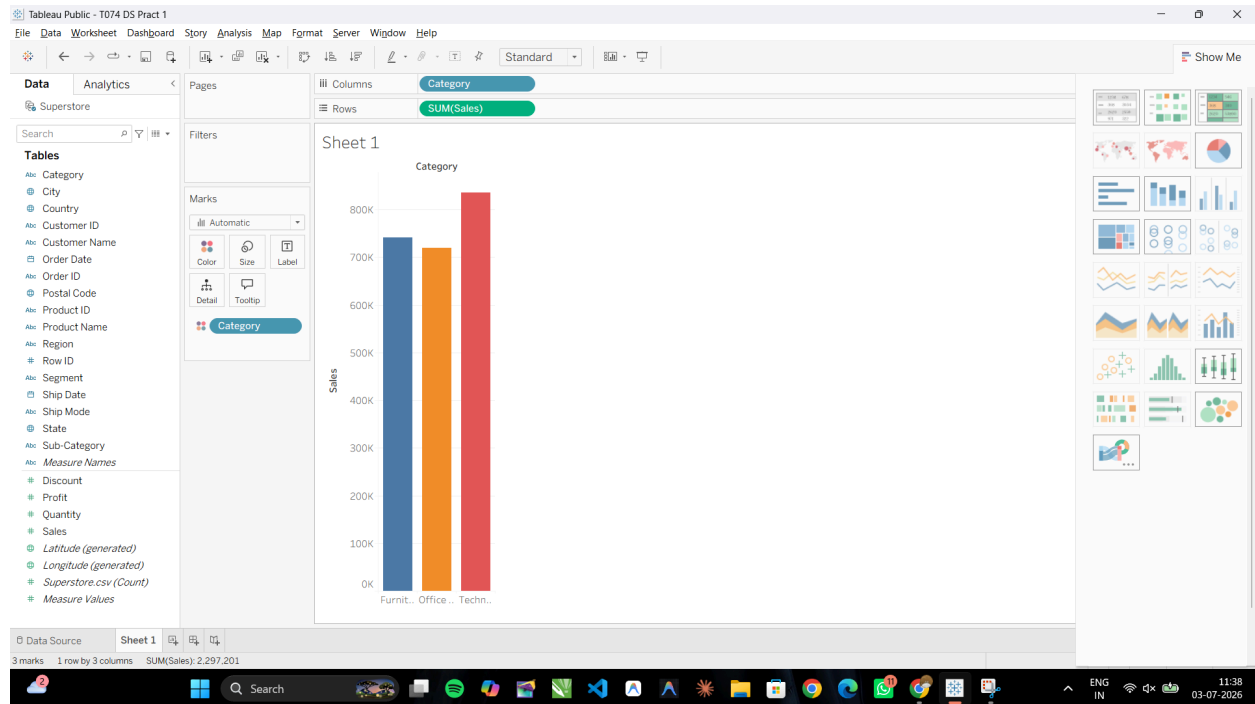
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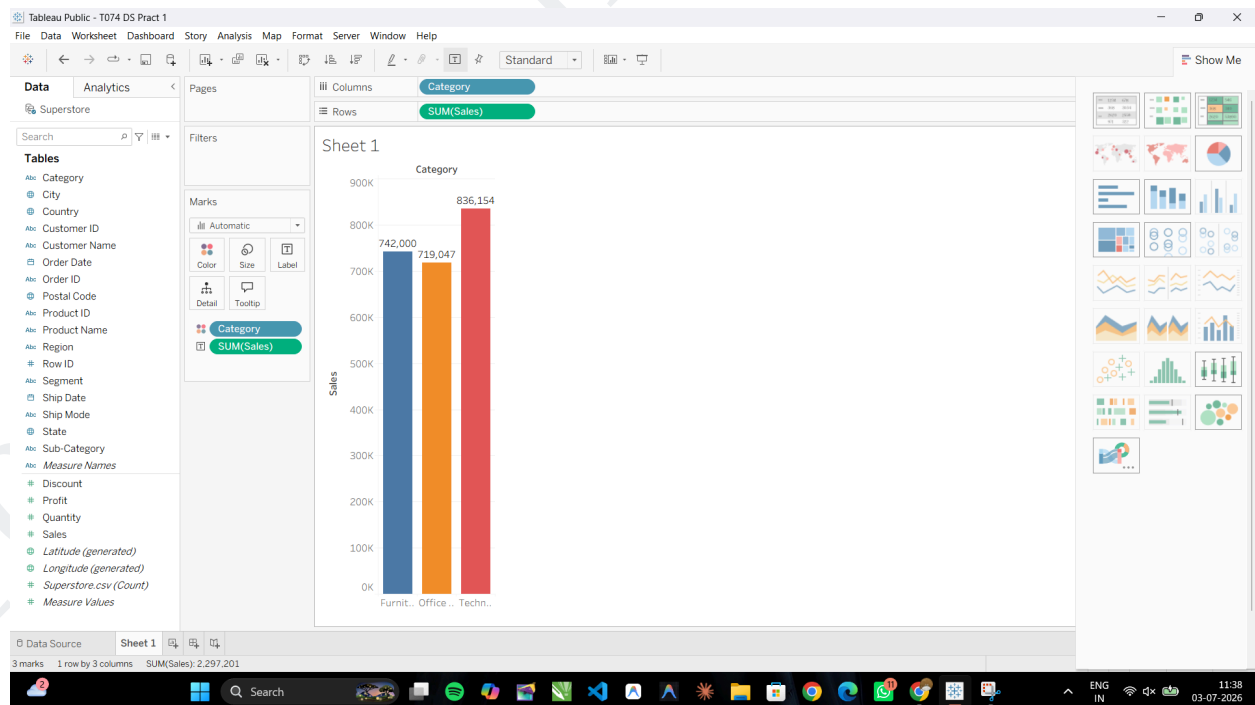
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3. Color customization: Take Category from the Data pane a second time, and drop it onto the Color tile in the Marks card



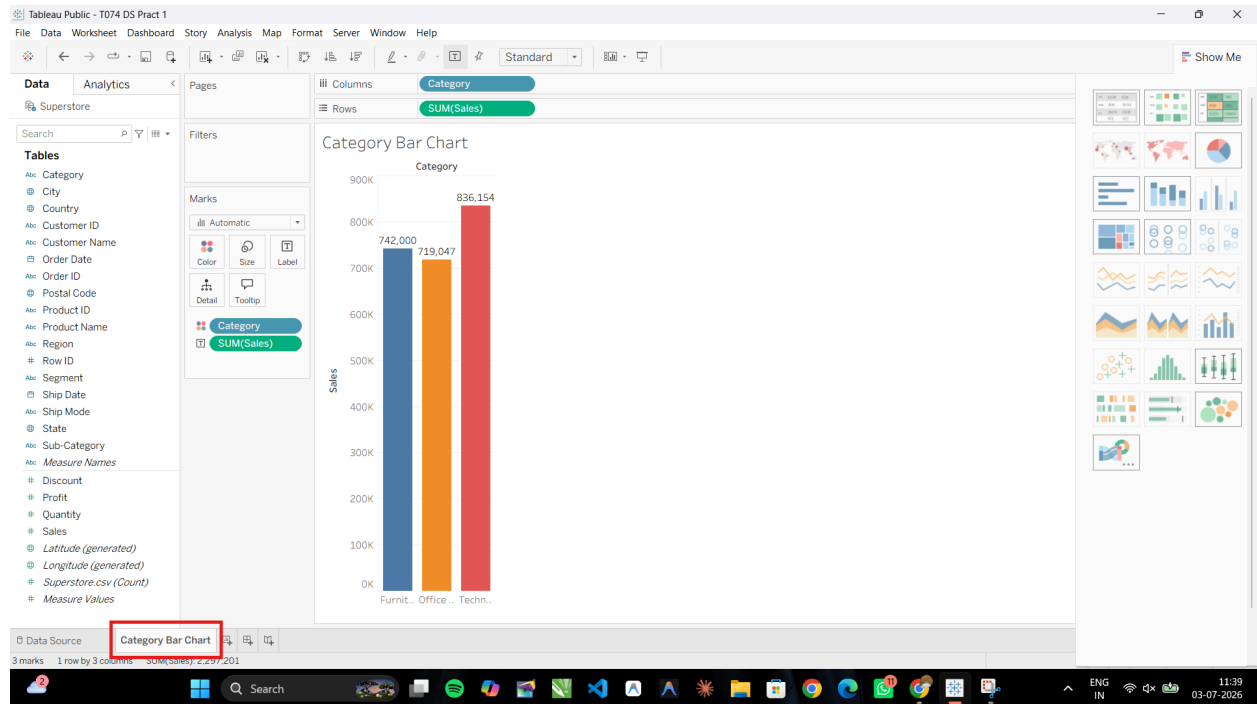
4. Label customization: Drag the Sales pill from the data pane and drop it onto the Label tile in the Marks card to display the precise revenue numbers over the bars



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5. Double-click the bottom tab sheet name Sheet 1 and rename it to Category Bar Chart

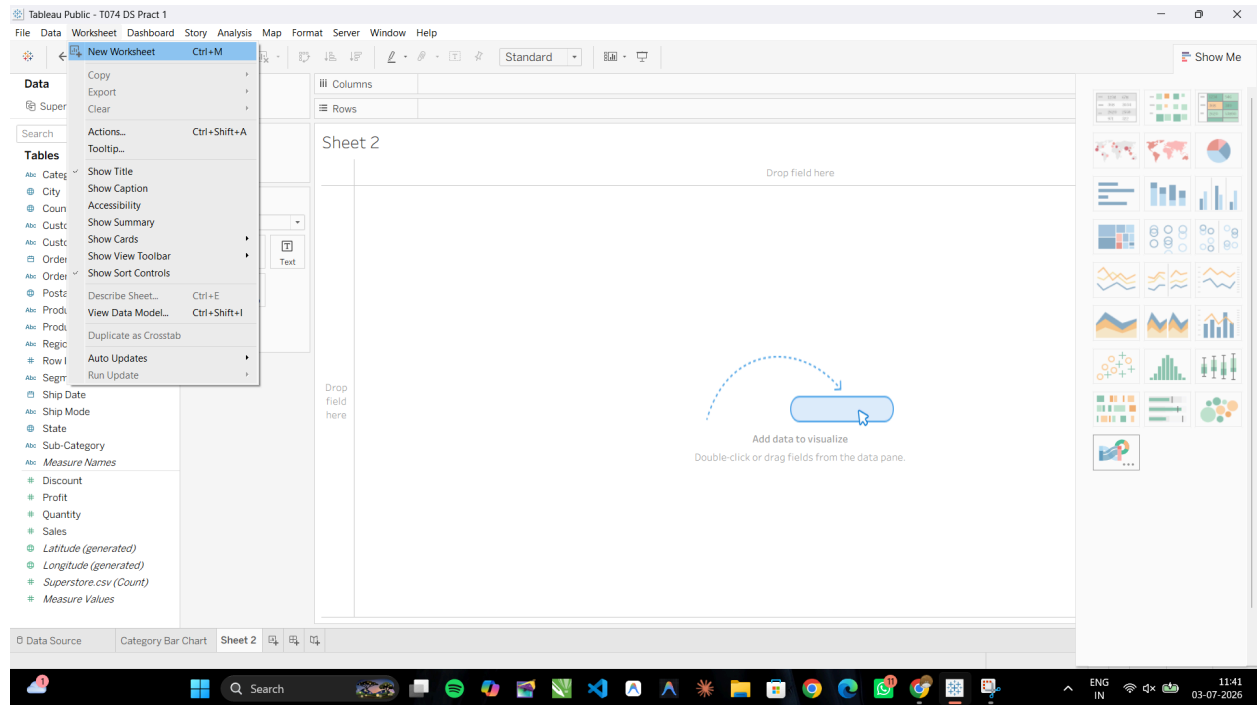


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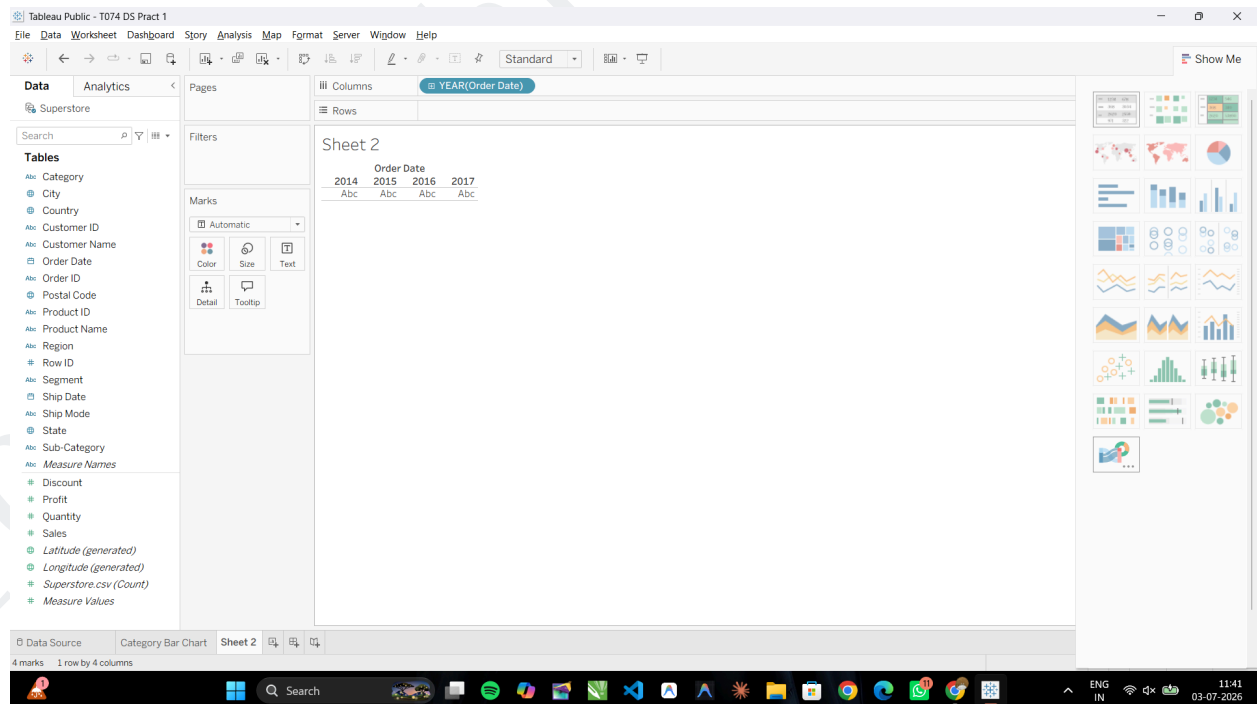
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Part E: Build a Line Chart for Time-Series Analysis

1. Click the New Worksheet button at the very bottom tab bar (the small icon with a chart and a plus symbol)



2. Drag the Order Date dimension onto your Columns shelf



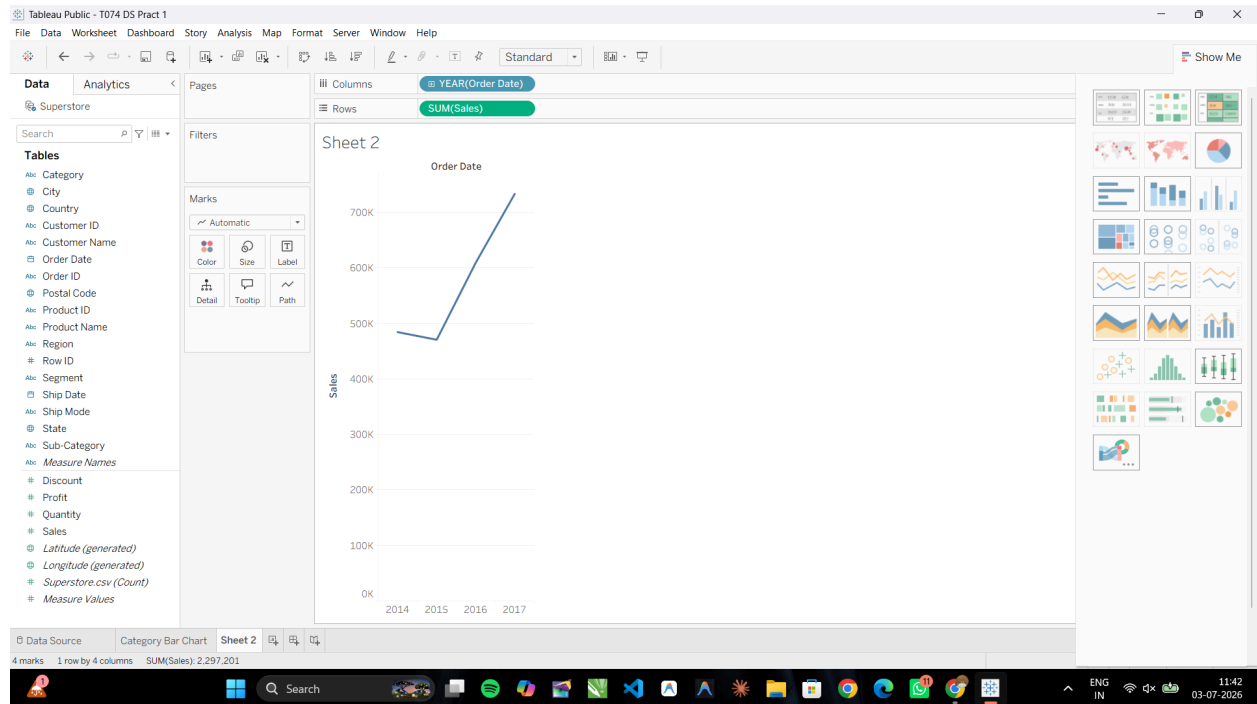
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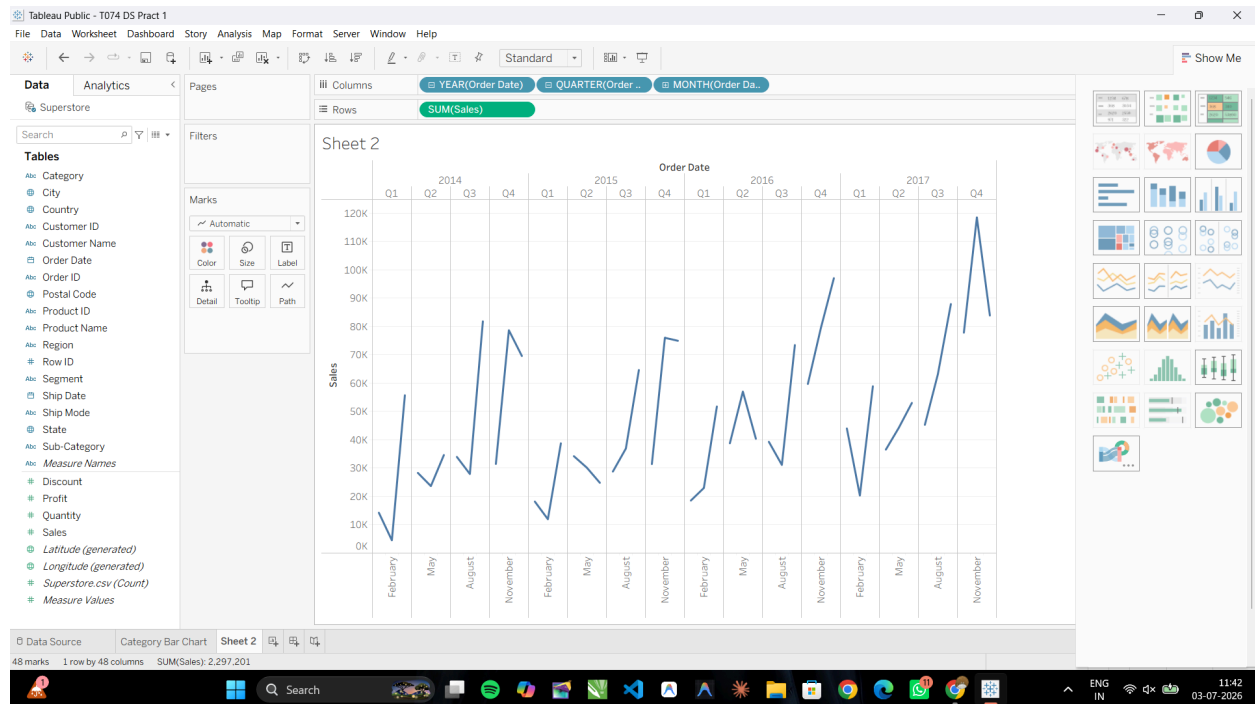
3. Drag the Sales measure onto your Rows shelf. Because a date field is on your columns shelf, Tableau defaults straight into a line chart



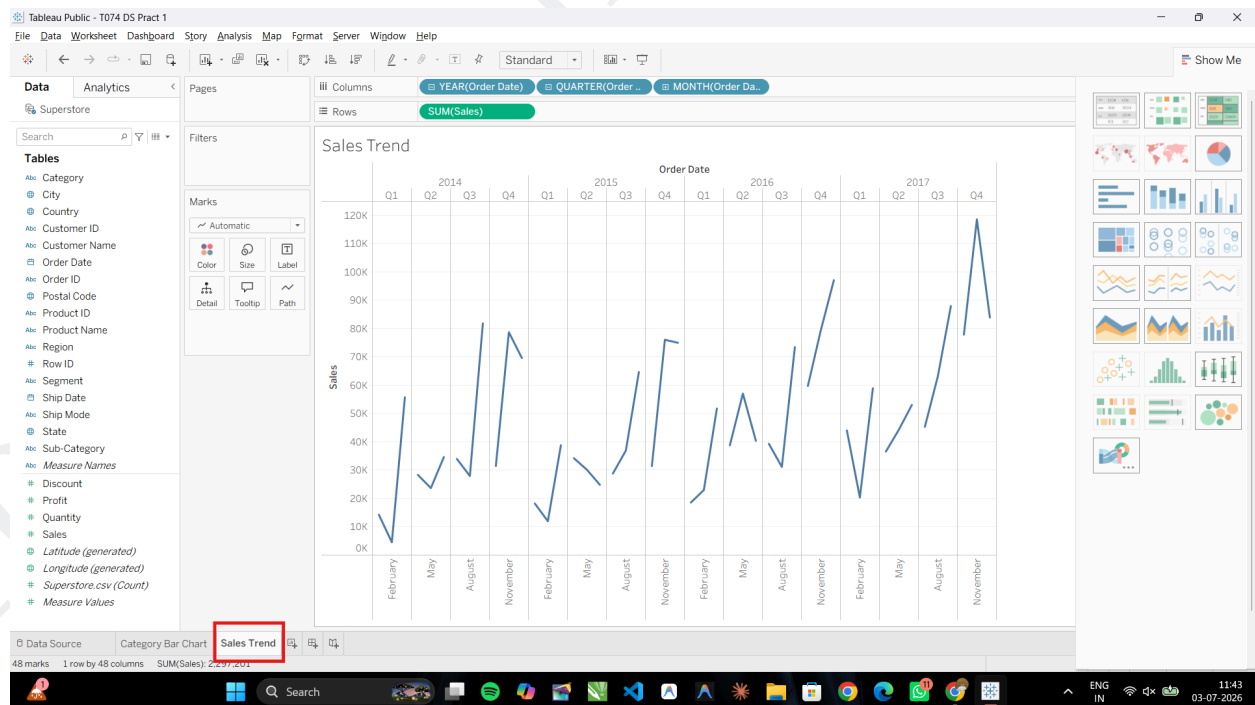
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4. Drilling Down: Look at the blue YEAR(Order Date) pill sitting in your Columns shelf. Click the small + icon located on that blue pill. It will expand your data into Quarters and Months, revealing hidden sales trends over time



5. Double-click the sheet tab at the bottom and rename this to Sales Trend

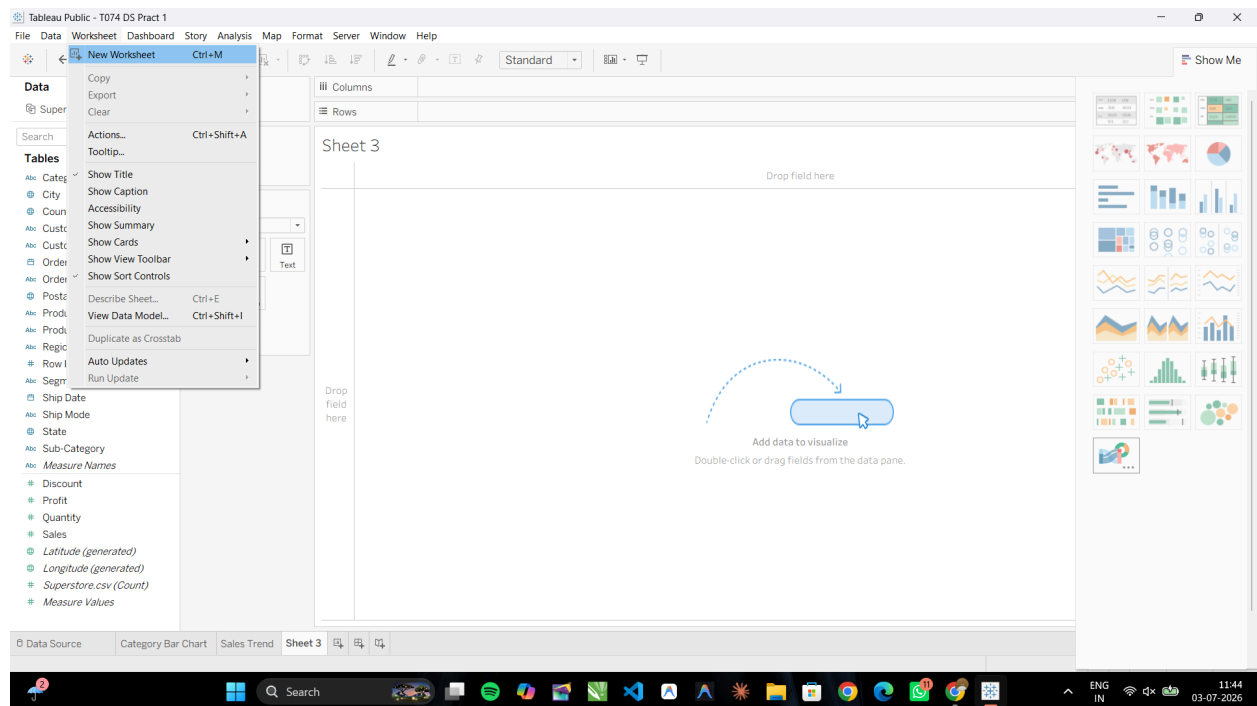


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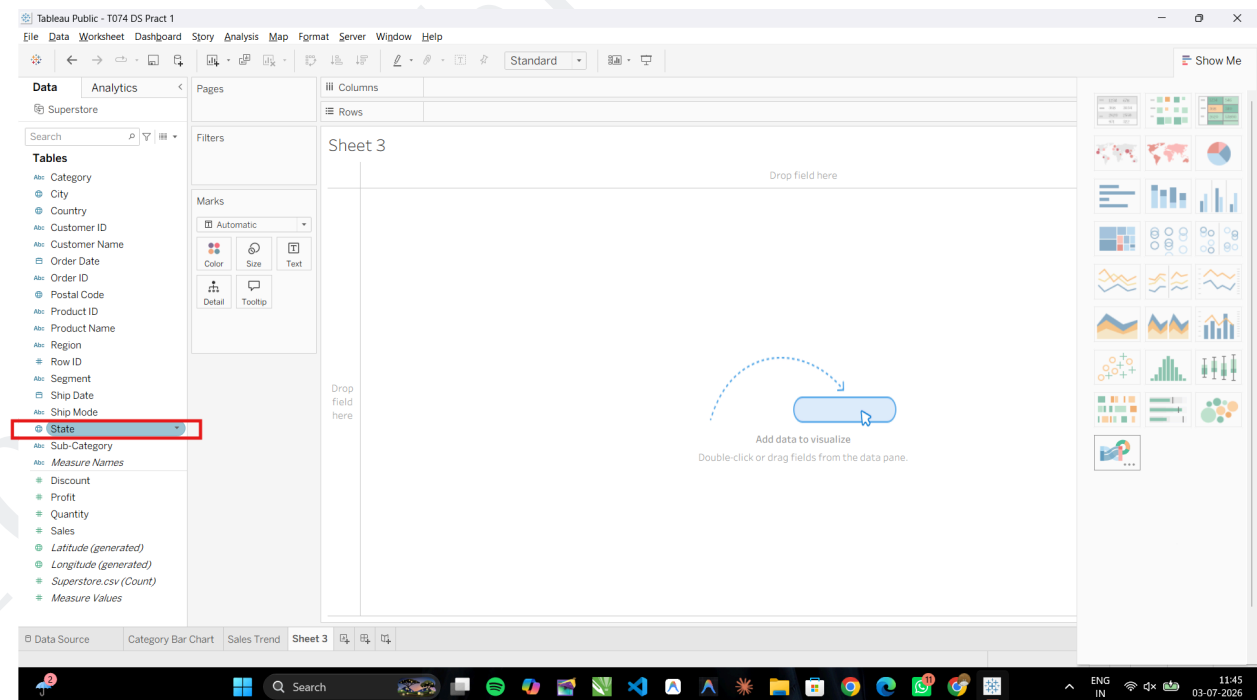
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Part F: Create a Geographic Map Visualization

1. Click the New Worksheet button at the bottom to create your third sheet



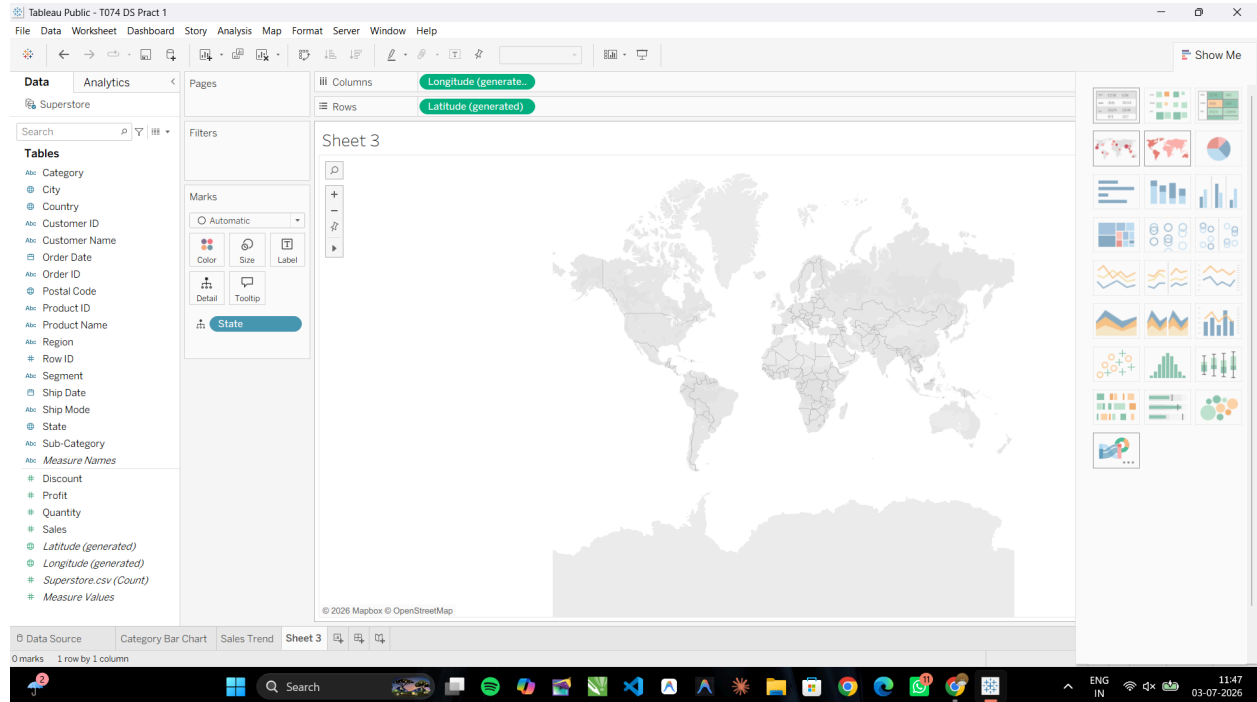
2. Look through your Data Pane and find the State field. It should display a tiny globe icon next to it, indicating geographic data



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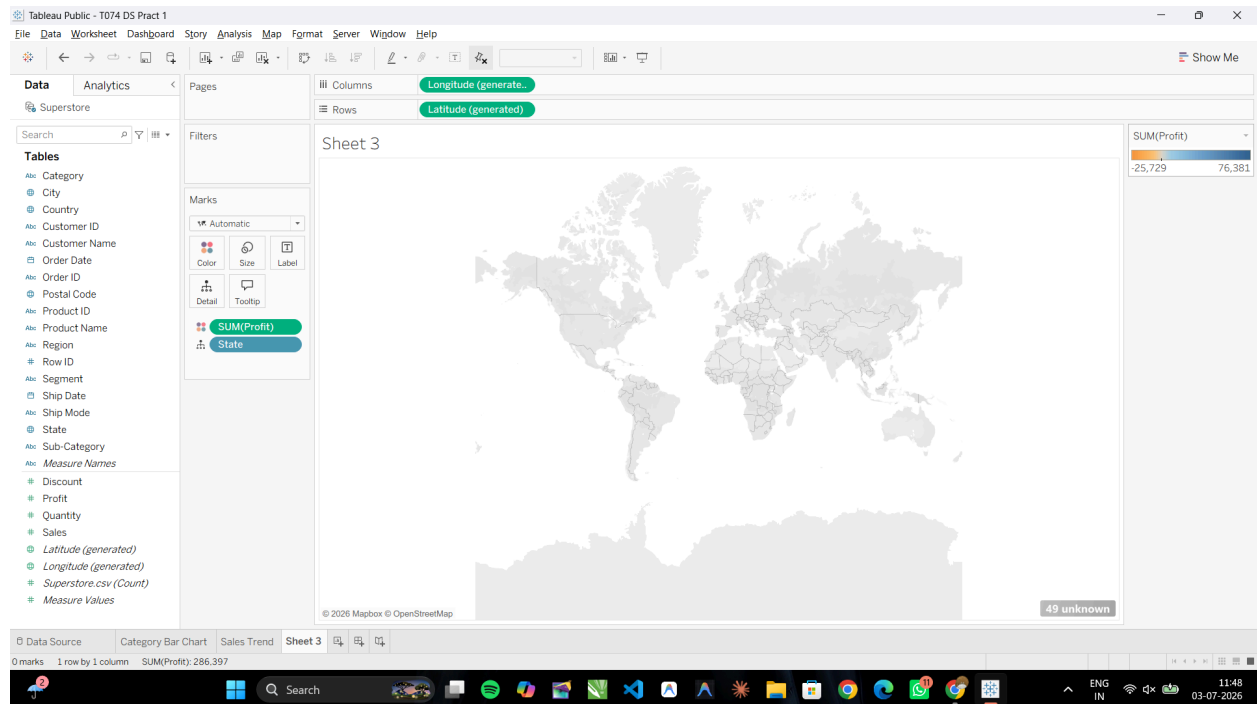
3. Double-click the State field. Tableau will automatically inject Latitude and Longitude coordinates into your shelves and draw a map



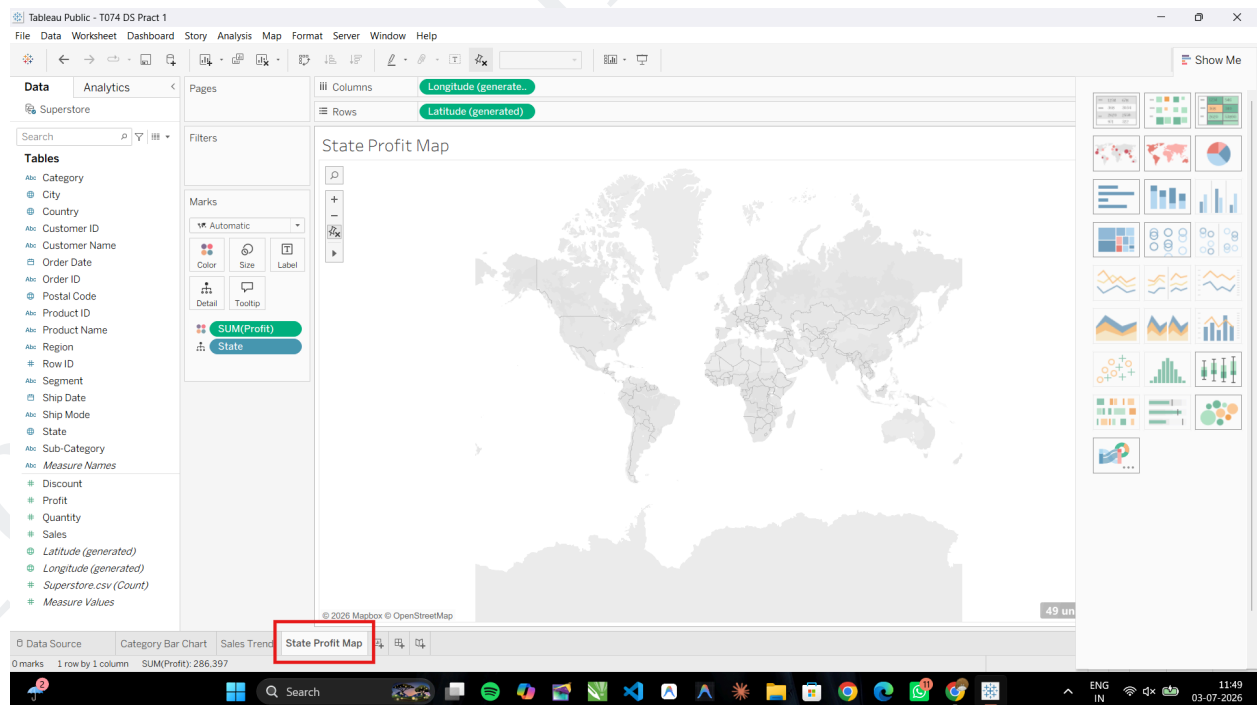
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4. Styling the Map: Go to your Data Pane, pick up Profit, and drop it directly onto the Color icon inside your Marks card. This will instantly color-code each state based on its performance



5. Rename this worksheet tab at the bottom to State Profit Map

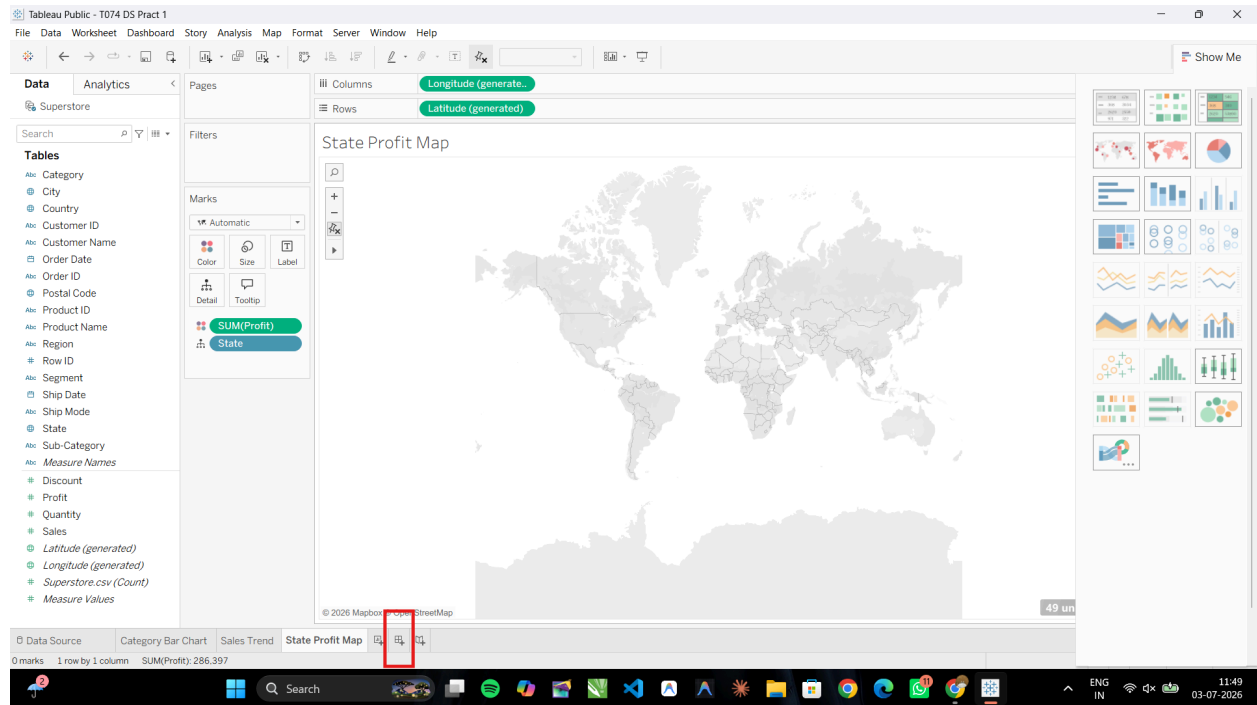


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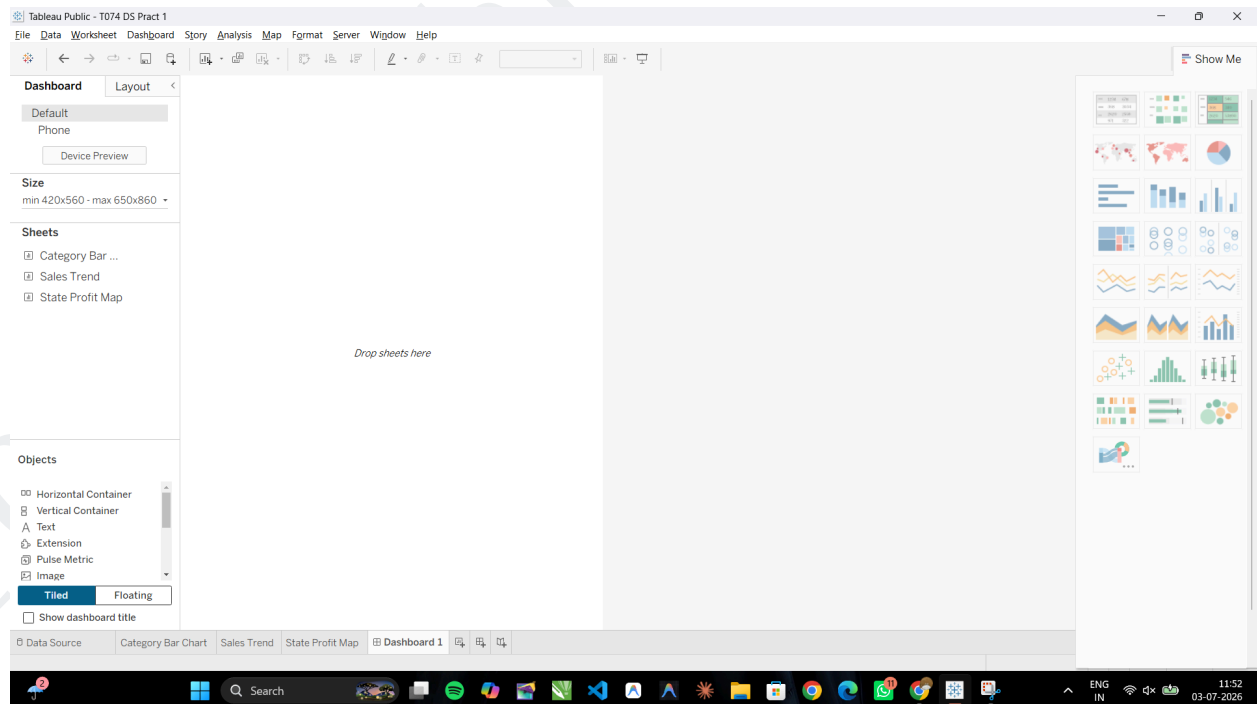
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Part G: Combine Charts into an Interactive Dashboard

1. Look at the bottom navigation tab bar and click the New Dashboard icon (it looks like a small 2x2 grid window pane with a plus symbol).



2. On the left side panel, you will see a list of your worksheets:



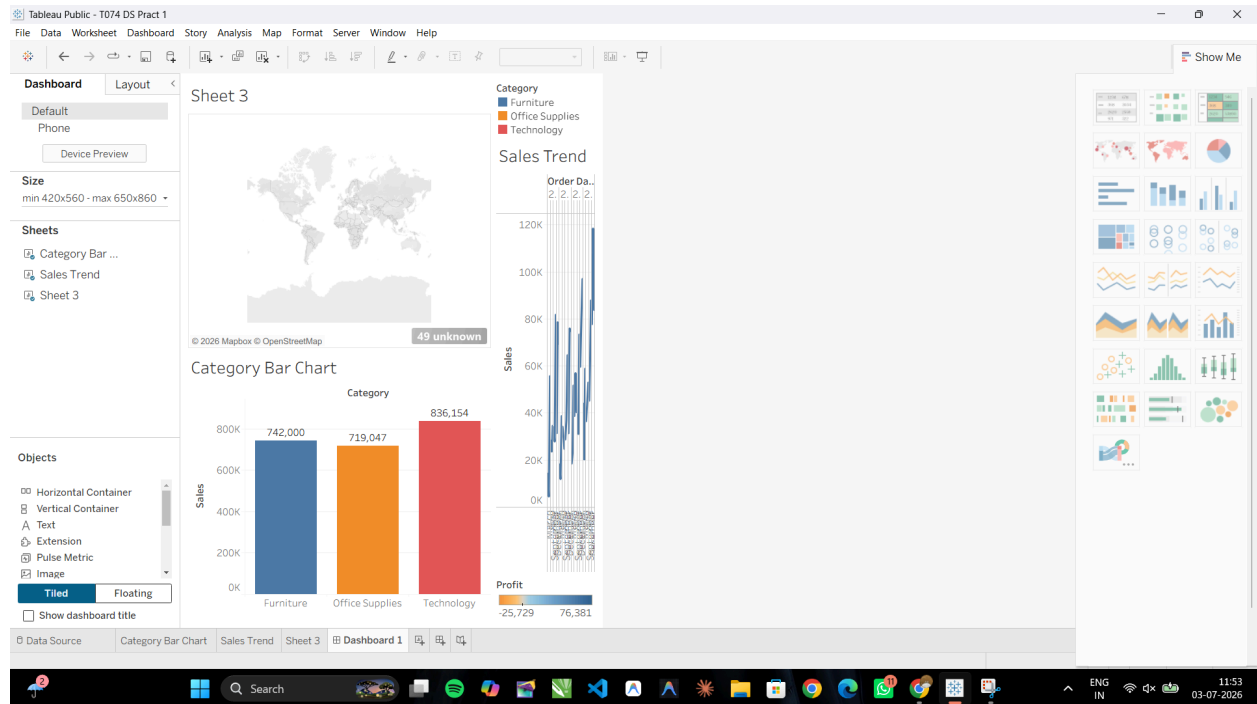
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3. Drag your Category Bar Chart, Sales Trend, State Profit Map and drop it alongside the right side of the layout.



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Turning on the Magic: Interactivity

1. Click once inside your Category Bar Chart zone on the dashboard to select it.
2. Look at the subtle gray menu options border that opens up on its top-right corner.
3. Click the tiny Funnel Icon (Use as Filter).

Now test out your creation! Click on the Technology bar in your bar chart. The map and the time-series line charts will dynamically isolate and update to display data exclusively for your Technology sales. Click the bar a second time to reset the filter.

